

Bootstrap Grid System

The Bootstrap Grid System is used to create responsive web page layouts. It divides the page into 12 columns. We can combine these columns to create different column widths.

Basic Structure

A Bootstrap grid generally contains:

Container – Holds the grid.

Row – Creates a horizontal group of columns.

Columns – Define how much space each element occupies.

Bootstrap 4 Grid Classes

Class	Screen Size	Description
.col-	Less than 576px	Extra small devices / phones
.col-sm-	≥ 576px	Small devices / tablets
.col-md-	≥ 768px	Medium devices
.col-lg-	≥ 992px	Large devices / desktops
.col-xl-	≥ 1200px	Extra-large desktops

Example 1: Three Equal Columns

```
<div class="container">  
  <div class="row">  
    <div class="col-md-4">Column 1</div>  
    <div class="col-md-4">Column 2</div>  
    <div class="col-md-4">Column 3</div>  
  </div>  
</div>
```

Example 2: Two Different Widths

```
<div class="container">
  <div class="row">
    <div class="col-md-8">Main Content</div>
    <div class="col-md-4">Sidebar</div>
  </div>
</div>
```

Example 3: Responsive Columns

```
<div class="container">
  <div class="row">
    <div class="col-12 col-md-8">
      Main Content
    </div>
    <div class="col-12 col-md-4">
      Sidebar
    </div>
  </div>
</div>
```

.....Bootstrap provides several predefined classes to create attractive, responsive, and well-formatted tables easily.

1. Basic Table

The `.table` class is used to add basic Bootstrap styling to a table. It provides padding and horizontal dividers.

Example:

```
<table class="table">
  <thead>
    <tr>
      <th>Name</th>
      <th>Course</th>
```

```
        <th>Marks</th>
    </tr>
</thead>
<tbody>
    <tr>
        <td>Rahul</td>
        <td>B.Sc IT</td>
        <td>85</td>
    </tr>
    <tr>
        <td>Priya</td>
        <td>B.Sc CS</td>
        <td>90</td>
    </tr>
</tbody>
</table>
```

2. Striped Rows Table

The `.table-striped` class adds alternate background colors to table rows, creating a zebra-stripe effect.

```
<table class="table table-striped">
    <tr>
    </tr>
    <tr>
        <td>Rahul</td>
        <td>85</td>
    </tr>
    <tr>
        <td>Priya</td>
        <td>90</td>
    </tr>
```

```
</table>
```

3. Bordered Table

The .table-bordered class adds borders around all sides of the table and its cells.

```
<table class="table table-bordered">
```

```
<tr>
```

```
<th>Name</th>
```

```
<th>Course</th>
```

```
</tr>
```

```
<tr>
```

```
<td>Rahul</td>
```

```
<td>B.Sc IT</td>
```

```
</tr>
```

```
</table>
```

4. Hover Rows Table

The .table-hover class adds a hover effect to table rows. When the mouse pointer moves over a row, its background changes.

```
<table class="table table-hover">
```

```
<tr>
```

```
<th>Name</th>
```

```
<th>Marks</th>
```

```
</tr>
```

```
<tr>
```

```
<td>Rahul</td>
```

```
<td>85</td>
```

```
</tr>
```

```
<tr>
```

```
<td>Priya</td>
```

```
<td>90</td>
```

```
</tr>
```

</table>

5. Contextual Classes

Contextual classes are used to give different meanings through colors to table rows or cells.

Class Meaning

.table-active Applies an active/grey highlighting

.table-success Indicates a successful or positive action

.table-info Indicates neutral or informative information

.table-warning Indicates a warning that needs attention

.table-danger Indicates a dangerous or negative action

```
<table class="table">
  <tr class="table-success">
    <td>Payment Successful</td>
  </tr>
  <tr class="table-warning">
    <td>Payment Pending</td>
  </tr>
  <tr class="table-danger">
    <td>Payment Failed</td>
  </tr>
</table>
```

6. Dark Table

The .table-dark class gives the table a dark/black background with light-colored text.

```
<table class="table table-dark">
  <tr>
    <th>Name</th>
    <th>Marks</th>
  </tr>
```

```

<tr>
  <td>Rahul</td>
  <td>85</td>
</tr>
</table>

```

7. Hoverable Dark Table

We can combine `.table-dark` and `.table-hover` to create a dark table with a hover effect.

```

<table class="table table-dark table-hover">
  <tr>
    <th>Name</th>
    <th>Course</th>
  </tr>
  <tr>
    <td>Rahul</td>
    <td>B.Sc IT</td>
  </tr>
  <tr>
    <td>Priya</td>
    <td>B.Sc CS</td>
  </tr>
</table>

```

8. Table Head Colors

Bootstrap 4 provides classes for changing the appearance of the table header.

`.thead-dark`

Creates a dark/black table header.

```

<table class="table">
  <thead class="thead-dark">
    <tr>

```

```

        <th>Name</th>
        <th>Marks</th>
    </tr>
</thead>
</table>


```

Creates a light/grey table header.

```

<table class="table">
    <thead class="thead-light">
        <tr>
            <th>Name</th>
            <th>Marks</th>
        </tr>
    </thead>
</table>

```

.....

Bootstrap Forms

Bootstrap provides predefined classes to create responsive and attractive forms easily. Forms are commonly used to collect information such as name, email, password, address, etc.

Types of Bootstrap Forms

There are mainly three types of form layouts:

Vertical Form – Default form layout

Horizontal Form – Labels and controls are arranged horizontally

Inline Form – Form elements are displayed in a single line

1. Vertical Form

In a vertical form, labels and form controls are placed one below another. It is the default form layout.

Example:

```

<form>

```

```
<div class="form-group">
  <label>Name:</label>
  <input type="text" class="form-control">
</div>
<div class="form-group">
  <label>Email:</label>
  <input type="email" class="form-control">
</div>
<button type="submit" class="btn btn-primary">Submit</button>
</form>
```

2. Bootstrap Inline Form

In an inline form, form controls are displayed side by side in a single line.

It is useful for small forms such as login, search, or subscription forms.

```
<form class="form-inline">
  <label class="mr-2">Name:</label>
  <input type="text" class="form-control mr-2">
  <label class="mr-2">Email:</label>
  <input type="email" class="form-control mr-2">
  <button type="submit" class="btn btn-primary">Submit</button>
</form>
```

3. Horizontal Form

In a horizontal form, the label and input field appear side by side.

In Bootstrap 4, the grid system is commonly used to create horizontal forms.

```
<form>
  <div class="form-group row">
    <label class="col-sm-2 col-form-label">Name:</label>
```

```

<div class="col-sm-10">
    <input type="text" class="form-control">
</div>
</div>
<div class="form-group row">
    <label class="col-sm-2 col-form-label">Email:</label>
<div class="col-sm-10">
    <input type="email" class="form-control">
</div>
</div>
<div class="form-group row">
    <div class="col-sm-10 offset-sm-2">
        <button type="submit" class="btn btn-primary">
            Submit
        </button>
    </div>
</div>
</form>

```

2. Use .form-group

The .form-group class provides proper spacing between form controls.

```

<div class="form-group">
    <label>Name:</label>
    <input type="text" class="form-control">
</div>

```

3. Use .form-control

The .form-control class is used with textual <input>, <textarea>, and <select> elements.

```

<input type="text" class="form-control">
<textarea class="form-control"></textarea>
<select class="form-control">

```

```
<option>India</option>
<option>USA</option>
</select>
```

Bootstrap 4 Form Controls

Bootstrap 4 automatically provides styling to form controls when the `.form-control` class is used.

Textual `<input>`, `<textarea>`, and `<select>` elements with `.form-control` generally take 100% width of their parent container.

Example:

```
<input type="text" class="form-control" placeholder="Enter your name">
```

Stacked / Full-Width Form

A stacked form displays form elements one below another and uses the available width of the parent container.

Example:

```
<form>
  <div class="form-group">
    <label>Username:</label>
    <input type="text" class="form-control"
      placeholder="Enter username">
  </div>
  <div class="form-group">
    <label>Password:</label>
    <input type="password" class="form-control"
      placeholder="Enter password">
  </div>
  <button type="submit" class="btn btn-success">
    Login
  </button>
</form>
```